Technical Club Portal — Project Report

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Course/Section: Btech CSE(aiml), 6th sem.
Live Site: https://p-kaizoku.github.io/cse_tech_club/

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1. Site Structure

The project is structured into four distinct HTML pages, organized to separate content, styling, and functionality (separation of concerns):

```
wt_assignment/
├── index.html          # Hero section, club mission, features, stat counters
├── events.html         # Dynamic listing of JS-rendered events
├── team.html           # Grid layout of core team members
├── contact.html        # Real-time validated registration form
├── css/
│   └── style.css       # Global design system & component styles
├── js/
│   └── main.js         # Interactive behavior and form validation
└── ss/                 # Screenshot directory
```

- **Semantic HTML5:** Applied `<header>`, `<nav>`, `<main>`, `<section>`, and `<article>` to provide meaning to algorithms and screen readers.
- **Shared Assets:** All pages share a single `style.css` file and `main.js` file, ensuring caching efficiency and absolute consistency across pages.

2. Key Design Decisions

The club portal was built using a **Monochrome / Brutalist / Editorial Design System** without any external libraries.

- **Monochrome Palette & Typography:** Instead of generic bright colors, a strictly black `#0a0a0a`, grey, and white palette is used paired with the bold font *Space Grotesk*. This gives the club a serious, premium, and tech-forward "hacker" identity.
- **Sharp Brutalist Elements:** Boxy borders (`1px solid #2e2e2e`) and sharp corners (`border-radius: 0`) were explicitly chosen over common rounded soft UI to invoke a structural, architectural feel. Grid gap trick (`gap: 1px` with a border background) was used to render crisp hairline dividers.
- **Responsive Layout Architecture:**
  - Designed mobile-first, using CSS Grid primarily for page layouts, transitioning from 1 column on mobile to 2/3/4 columns on desktop using `@media (max-width: 1024px)` and `@media (max-width: 600px)` breakpoints.
  - Flexbox is used for 1-dimensional components (Navigation, footer alignments, form groups).
- **JavaScript Enhancements:**
  - **Dark/Light Theme Toggle:** A custom robust toggle using `localStorage` ensures user preference is saved, mapping directly to CSS Variables.
  - **Dynamic Rendering:** Event cards are generated by iterating an array of JSON objects, simulating a real data-fetching experience.
  - **Client-Side Validation:** Form inputs give realtime feedback on `blur` and `submit`, using Regex patterns for the college email and 10-digit phone number.

- **Scroll Reveal Animations:** IntersectionObservers smoothly fade elements in as they enter the viewport, giving a polished feel.

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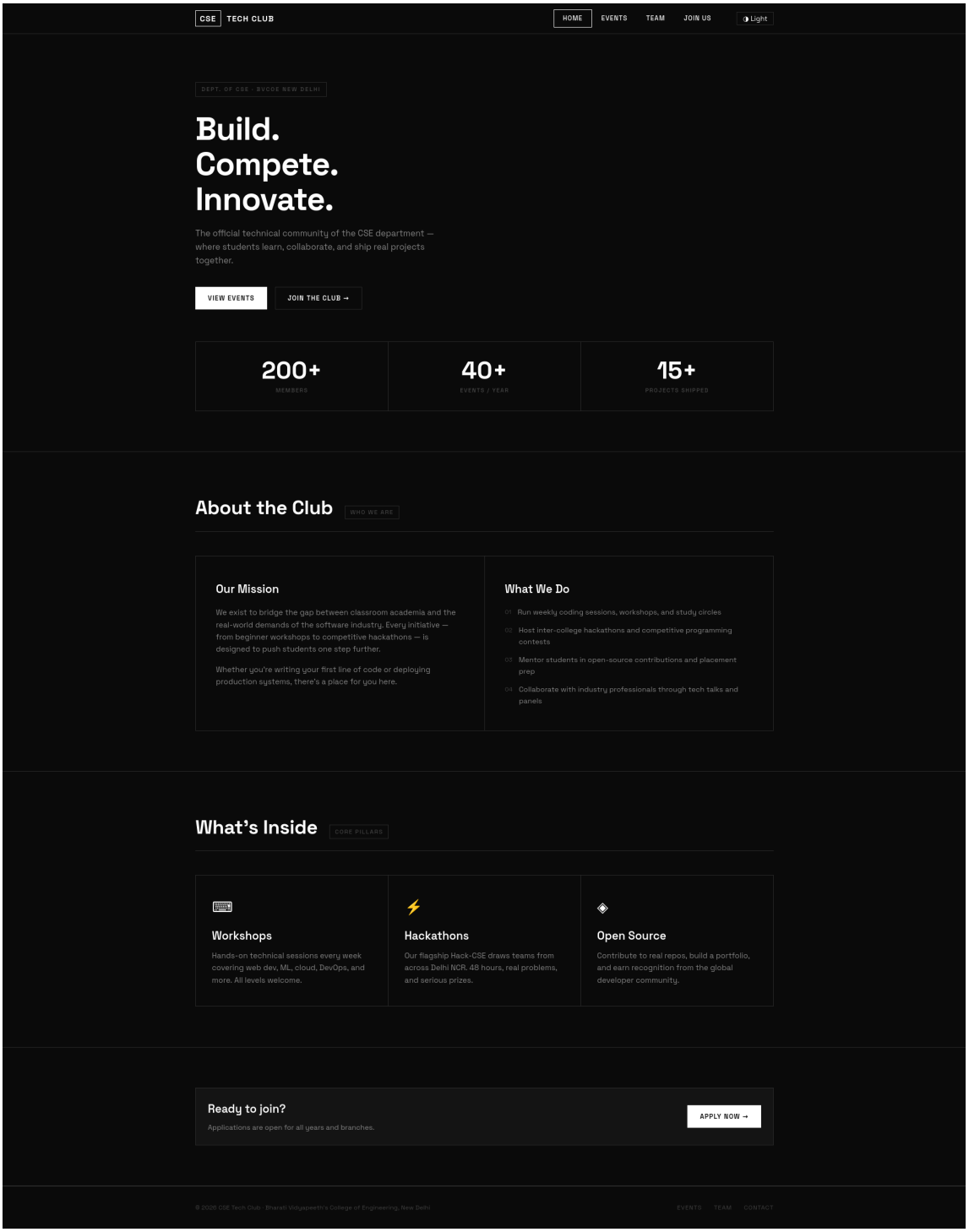
3. Page Screenshots

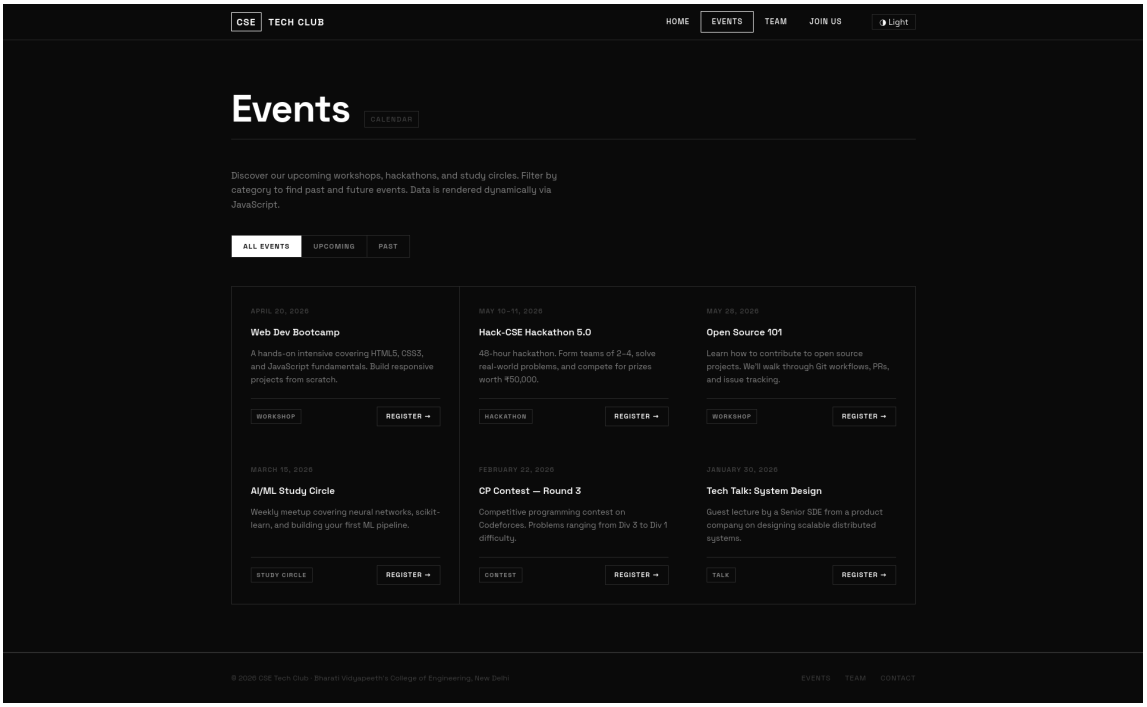
Home Page

Features the Hero banner, about section, and feature cards.



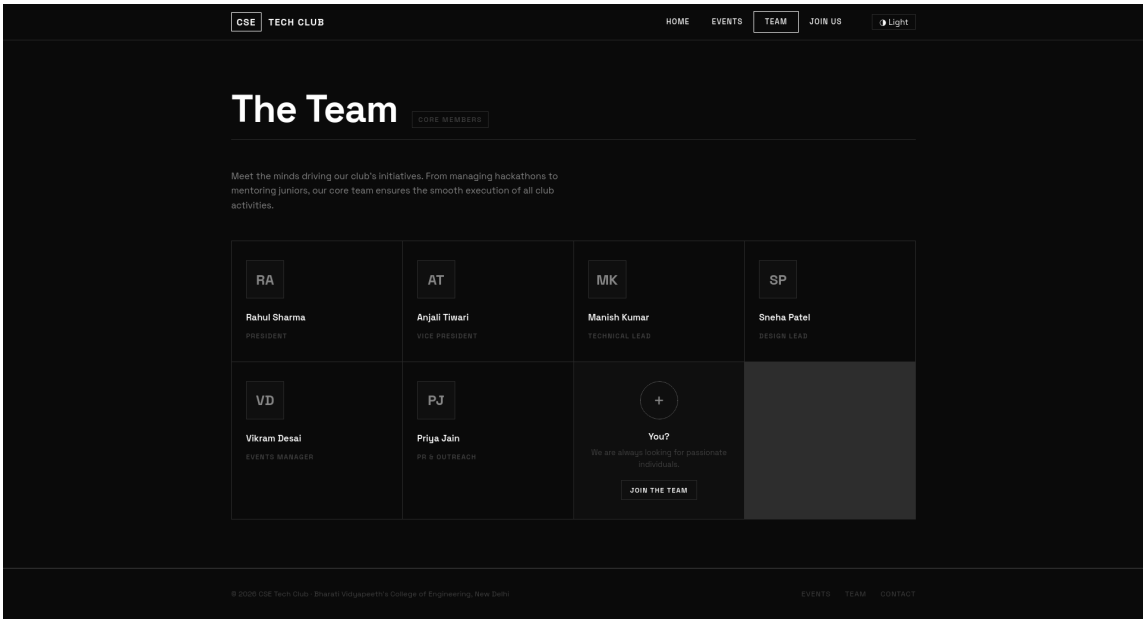
Events Page

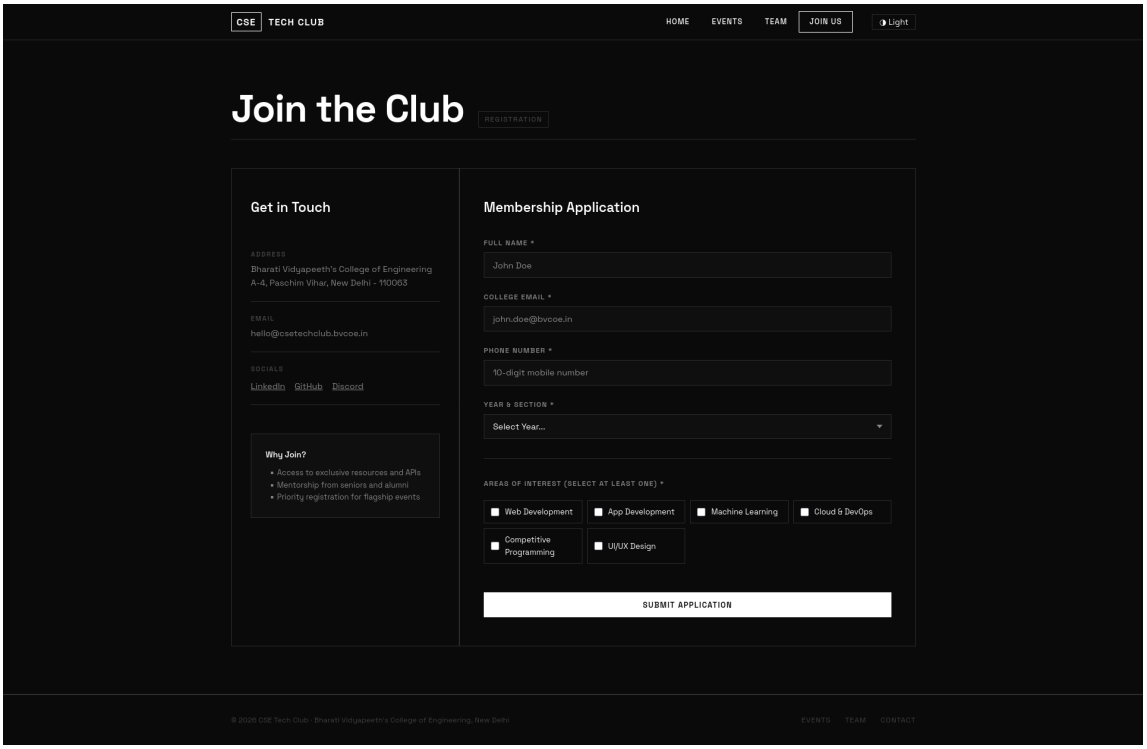
Dynamically populated list of events and hackathons.



Team Page

Responsive grid showing member cards with hover-bio effects.



Contact / Registration Page

Realtime validation form for club applications.